STREAK IT — COMPETITOR RESEARCH

VIBL + PraxiLabs | Microbiology Virtual Laboratory Competitive Analysis

Purpose: Consolidated competitor findings, experiment overlap, additional experiment coverage, and potential differentiation opportunities for Streak It.

1. Competitors Compared

VIBL — Virtual Interactive Bacteriology Laboratory

VIBL is a Michigan State University virtual bacteriology laboratory focused on experiment-based learning. Main sections include Gram Stain, Streak Plate, Differential Media, Biochemical Tests, and Kirby-Bauer.

Observed workflow: the learner opens an experiment, reads a short description, clicks the module, sees Description, Steps, Credits, and Start, and then begins. Equipment can be dragged and dropped. Incorrect actions trigger notifications.

Primary focus: experiments and their instructions rather than broader skill-development or conversational learning.

PraxiLabs

PraxiLabs is a broad 3D virtual STEM laboratory platform covering Biology, Microbiology, Molecular Biology, Biochemistry, Genetics, Chemistry, Physics, and Microscopy.

Observed features include a large experiment library, interactive simulations, free-trial access, reports, quizzes, performance tracking, LMS-related functionality, learning resources/e-books, and institutional features.

2. Important Correction: PraxiLabs' Experiment Bot

PraxiLabs should not be described as having a standalone conversational chatbot. During an experiment, a bot/icon appears and can be clicked to open an instruction box. Some experiments contain instructions in this box, while others may show an empty instruction area.

This differs from a dedicated Streak It chatbot that could support conversational questions, explanations, hints, troubleshooting, and contextual learning.

3. LMS Integration — Research Observation

PraxiLabs advertises LMS integration for institutional/instructor use. Based on the observed experience, this should not be represented as a complete LMS replacement. The integration is better understood as an institutional/instructor connection, including use with systems such as Blackboard.

Streak It should therefore avoid claiming differentiation merely because it has LMS integration. A stronger opportunity is a more integrated student learning environment combining experiments, chatbot support, progress, assessment, and learning resources.

4. Experiments Common to VIBL and PraxiLabs

Area	*Competitor Evidence*	*Streak It Opportunity*
Gram Stain	*VIBL: Gram Stain; PraxiLabs: Gram Stain Simulation*	*Advanced interactive procedure, AI guidance, reagent explanations, result interpretation and mastery tracking.*
Streak Plate	*VIBL: Streak Plate; PraxiLabs: Bacterial Plating-Out Technique — Streak Plate Simulation*	*Practice, challenge and assessment modes; procedural scoring and feedback.*
Biochemical Testing	*VIBL: Biochemical Tests; PraxiLabs: multiple biochemical tests*	*Connect individual tests into a bacterial-identification pathway.*
Kirby-Bauer / Antibiotic Sensitivity	*VIBL: Kirby-Bauer; PraxiLabs: Disc Diffusion Method*	*Measure/interpret inhibition zones and make evidence-based decisions.*
Differential Media	*VIBL: Differential Media; PraxiLabs: related culture/biochemical experiments*	*Media interpretation and guided identification activities.*

5. Common Experiment Summary

Feature	*VIBL*	*PraxiLabs*	*Streak It Opportunity*

Gram Stain	*Yes*	*Yes*	*Advanced interactive + AI guidance*
Streak Plate	*Yes*	*Yes*	*Practice → Challenge → Mastery*
Biochemical Tests	*Yes*	*Yes*	*Diagnostic pathway*
Kirby-Bauer	*Yes*	*Yes*	*Interpretation + decision-making*
Differential Media	*Yes*	*Related concepts*	*Media interpretation*
Microbial Identification	*Related*	*Related*	*Potential major opportunity*
Experiment Instructions	*Yes*	*Yes / varies*	*Context-aware instructions + chatbot*
Error Notification	*Yes*	*Interactive feedback*	*Detailed explanation of errors*
Standalone Chatbot	*No*	*No, based on testing*	*Potential major differentiator*
Progress Tracking	*Limited*	*Yes*	*Skill-based mastery*
Streak System	*No*	*Not a core feature*	*Core Streak It feature*
E-books / Resources	*Not observed*	*Yes*	*Integrated learning resources*

6. Additional Microbiology Experiments — PraxiLabs

Staining & Microscopy

· *Ziehl-Neelsen Stain Technique*

· *Virtual Microscopy Lab — Introduction to Microscopes*

Bacterial Identification / Biochemical Testing

· *Catalase Test*

· *Coagulase Test*

· *Oxidase Test*

· *Indole Microbiology Test*

· *Urease Test*

· *Citrate Utilization Test*

· *Triple Sugar Iron Agar Test*

Microbial Culture / Isolation

· *Bacterial Plating-Out Technique — Streak Plate Simulation*

Antibiotic Testing

· *Antibiotic Sensitivity Virtual Lab — Disc Diffusion Method*

Virology / Microbiology

· *Virus Simulator — Cultivation and Preparation of Virus in Chick Embryo*

Microbiology-Related Biological Testing

· *Haemagglutination Test*

· *Washed Red Blood Cells Preparation*

· *Polarographic Oxygen Analyzer Respirometry*

7. Additional Molecular Biology / Genetics Experiments — PraxiLabs

Molecular Biology

· *DNA Extraction*

· *RNA Extraction*

· *First Strand cDNA Synthesis*

· *Conventional PCR*

· *Real-Time PCR*

· *Agarose Gel Electrophoresis*

· *DNA Sequencing*

· *DNA Microarray*

Genetics / DNA Technology

· *DNA Cloning — Isolation and Restriction Digestion*

· *Cloning — Transformation*

· *Blue and White Screening*

· *Plasmid Growth and Extraction*

· *Plasmid DNA Gel Electrophoresis*

· *Plasmid DNA Isolation*

· *DNA Fingerprinting using RFLP*

Competitive significance: PraxiLabs competes strongly through breadth. Streak It can instead compete through microbiology depth, guidance, practice, and personalization.

8. PraxiLabs E-book / Learning Resource Facility

PraxiLabs also provides an e-book/learning-resource facility, extending the platform beyond simulation-only learning.

Potential Streak It improvement:

· *Before Experiment — reading*

· *Experiment — interactive procedure*

· *AI Lab Mentor — contextual questions and explanations*

· *Assessment — test understanding*

· *Report — performance and mistakes*

· *Recommendation — targeted reading or next experiment*

Integrated learning loop: Read → Simulate → Practice → Assess → Review.

9. Potential Streak It Differentiators

Dedicated Chatbot

A standalone microbiology AI chatbot available throughout the platform, rather than an icon that only opens an instruction panel.

Context-Aware AI

The chatbot understands the current experiment and can explain why a step is performed, what an error means, or what concept should be reviewed.

Streak + Skill Mastery

Track activity streaks and actual competencies such as staining, aseptic technique, culture techniques, biochemical testing, and antibiotic sensitivity.

Experiment-to-Experiment Learning Path

Connect experiments into logical diagnostic or learning pathways instead of treating every experiment as isolated.

Persistent Learning History

Store attempts, scores, common mistakes, mastery level, improvement over time, and recommended next activities.

10. Example: Streak It Microbiology Learning Path

Unknown bacterial sample

↓

Streak Plate

↓

Observe colonies

↓

Gram Stain

↓

Select an appropriate biochemical test

↓

Interpret results

↓

Select the next test

↓

Identify the organism

This approach makes students reason through evidence and laboratory results rather than only following a fixed procedure.

11. Recommended Competitor Research Framework

1. *Competitor Overview*
2. *Target Audience*
3. *Homepage & Positioning*
4. *Registration / Onboarding*
5. *Experiment Library*
6. *Common Experiments with Streak It*
7. *Additional Microbiology Experiments*
8. *Other Subject Areas*
9. *Experiment Interface*
10. *Instructions & Guidance*
11. *AI / Chatbot*
12. *Error Detection & Feedback*
13. *Assessment*
14. *Progress Tracking*
15. *Attempts / Reports*
16. *Gamification*
17. *Streak System*
18. *Learning Resources / E-books*
19. *LMS Integration*

20. Student Dashboard

21. Instructor Features

22. Institutional Features

23. Pricing / Free Trial

24. Strengths

25. Weaknesses

26. Potential Gaps

27. What Streak It Can Do Better

28. Potential Unique Features for Streak It

29. UX Observations

30. Final Competitive Verdict

12. Final Competitive Positioning

VIBL: a focused virtual bacteriology experiment platform centered on experiments, instructions, and interactive procedures.

PraxiLabs: a broad 3D virtual STEM laboratory platform focused on a large simulation library, assessments, reporting, resources, and institutional functionality.

***Potential Streak It positioning:** A microbiology-focused virtual learning environment combining interactive experiments, a dedicated AI lab mentor, skill-based progress tracking, streak-based practice, connected experiment pathways, and personalized learning.*

***Strategic principle:** Streak It should not compete only by claiming to have more experiments. A stronger proposition is: "We help students become better at microbiology."*